

SIMPLIFIED CONSTRUCTION ESTIMATE

Cement : $21 \times .563 = 11.82$ say 12 bags

Sand : $21 \times .0375 = .79$ cu. m.

Gravel : $21 \times .075 = 1.57$ cu. m.

Problem Exercise

1. By the Area Method, determine the quantity of cement, sand and gravel of the proposed concrete parking area 6 inches thick with a general dimensions of 150 x 120 meters using class "A" mixture.

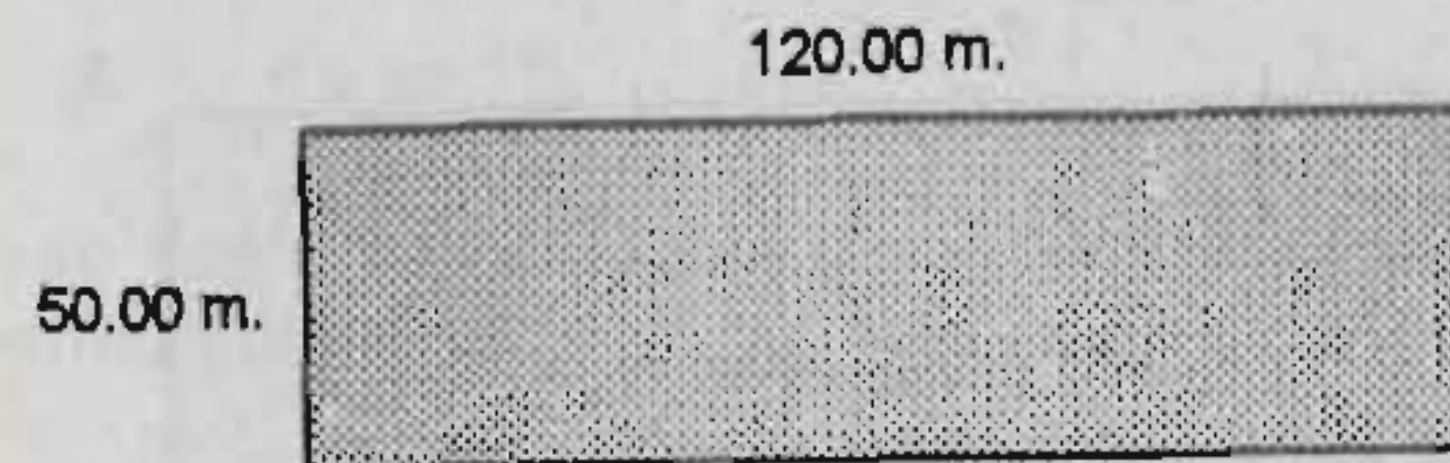


FIGURE 1-9 FLOOR PLAN OF PARKING AREA

2. By the Volume and Area Method, find the quantity of cement, sand and gravel of the concrete pavement 4" thick as shown in Figure 1-10 using class "B" concrete.

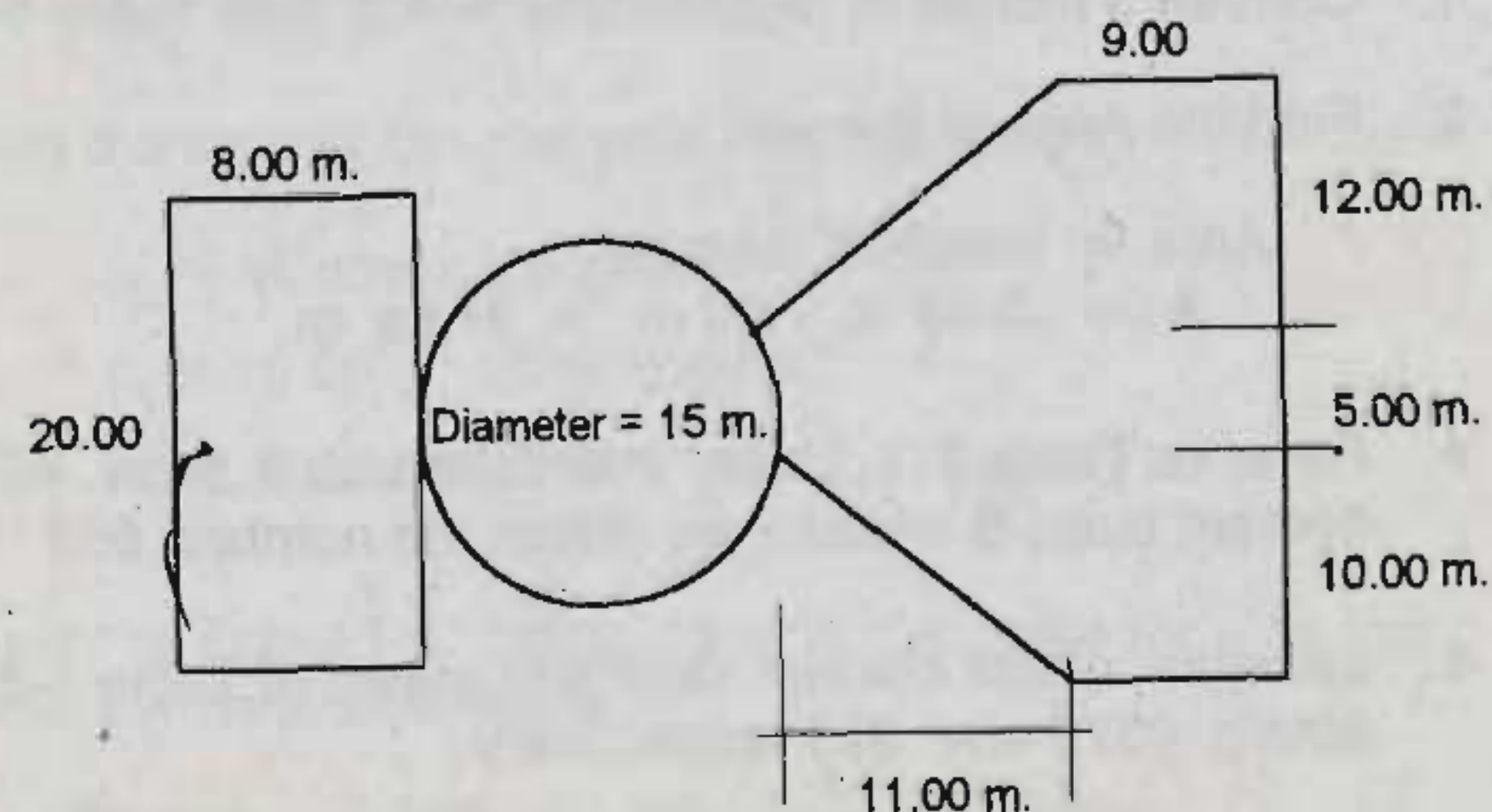


FIGURE 1-10 CONCRETE PAVEMENT PLAN

CONCRETE

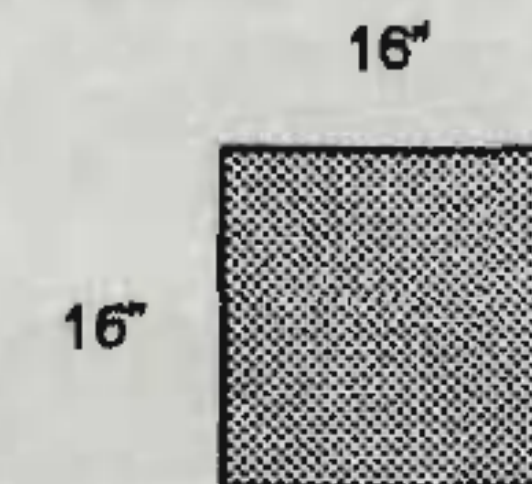
1-7 SQUARE CONCRETE COLUMN

Estimating the quantity of materials for concrete post or column can also be done in two simple ways:

1. By Volume Method and
2. By Linear Meter Method.

ILLUSTRATION 1-7

A reinforced concrete square column is 5.00 meters long with a cross sectional dimensions of 16 by 16 inches. Determine the quantity of cement, sand and gravel using class A concrete.



Cross Section

FIGURE 1-11 SQUARE COLUMN

SOLUTION :

A. By Volume Method

1. Convert first all measurements from inches to meter. 16 inches = .40 meter (See Table 1-1)
2. Solve for the cross sectional area of the column.

$$\text{Area : } .40 \times .40 = .16 \text{ sq. m.}$$

3. Find the volume of the column.